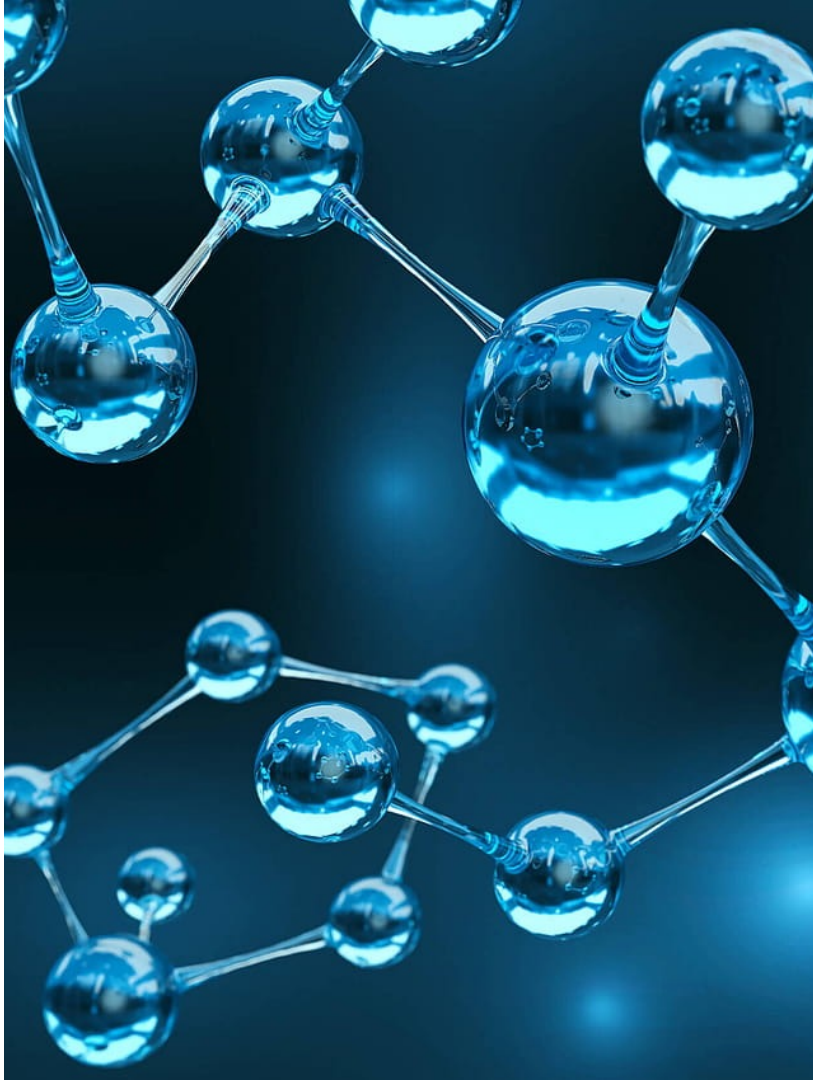
Mutexa

Digital Enzymology Wet Lab

Since 2023

mutexa@gmail.com



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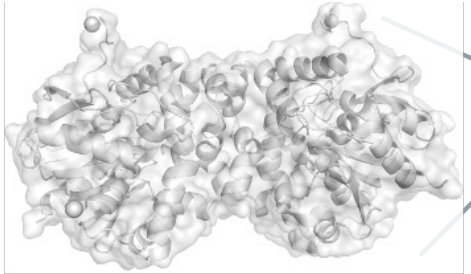
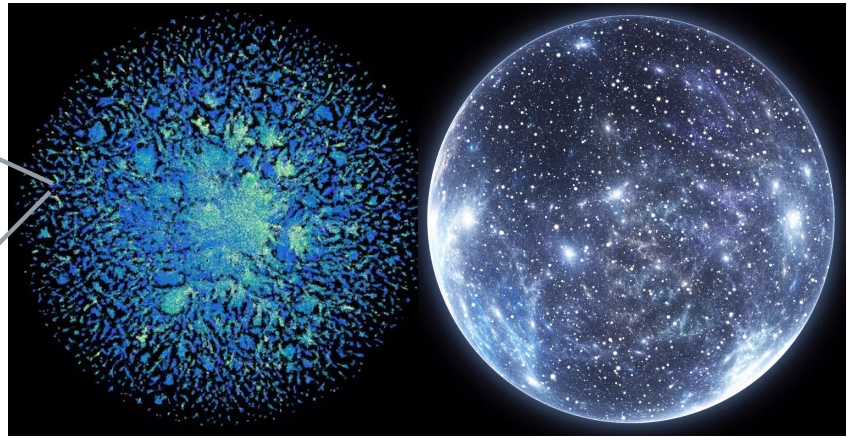
Challenges - Enzyme Engineering Life Cycle

Direct Evolution → Expression → Purification → Assay



Antibiotic Synthase



Amino Acid Sequence Space >> Stars in the Universe

- Time-consuming
- Labor-intensive
- Limited success rate
- Constrained sequence space exploration

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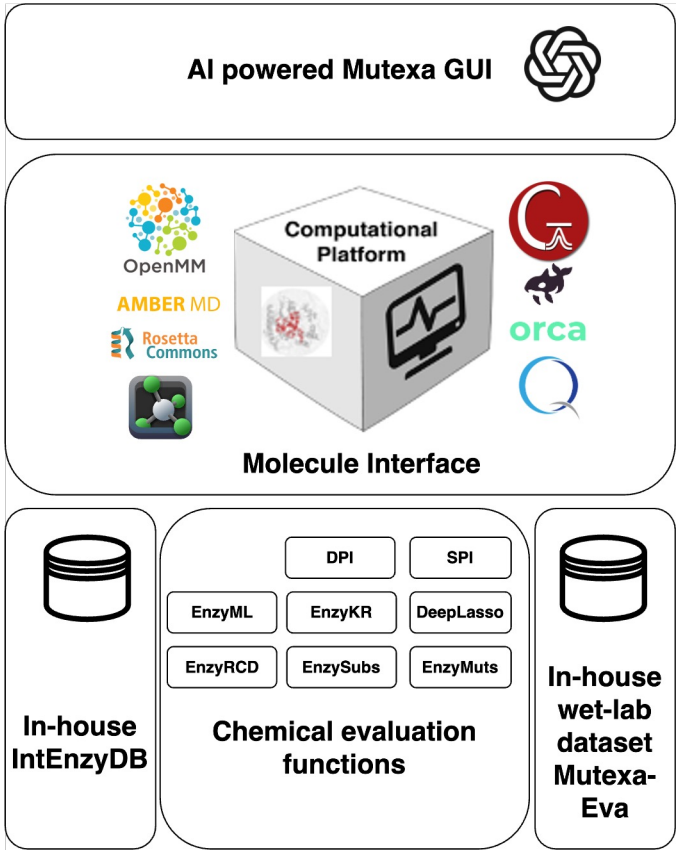
Our Solution - AI Powered Digital Wet Lab

- Taking on the status-quo with Mutexa:
- software powered by simulation and AI

- Functions like a self-driving car simulator for enzyme
- discovery

- Leveraging GPT and LLMs for an
- easy-to-use tool

- Users simply input their desired enzyme properties via a prompt - the backend takes care of the rest!



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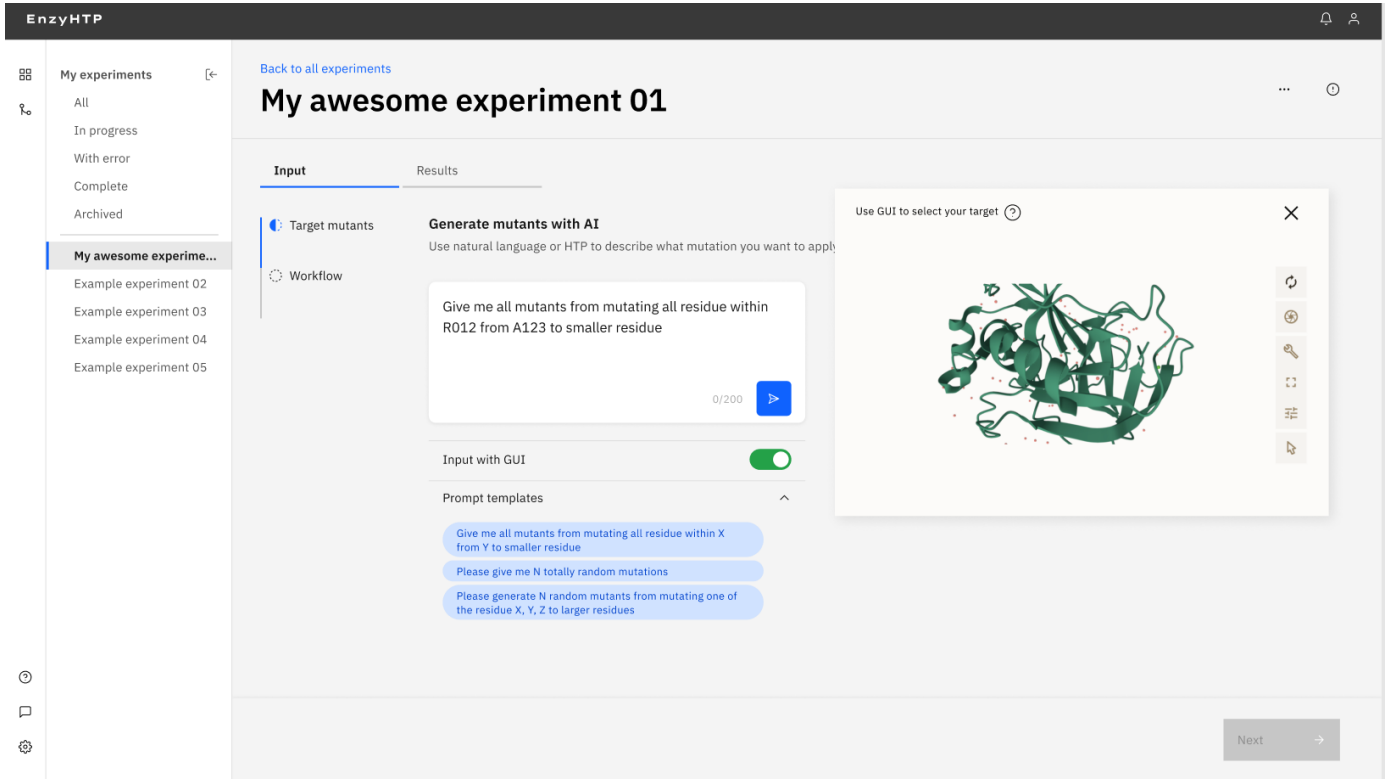
Product - Mutexa User Interface



AI Agent integrated platform will assist beginner to work with the simulation softwares, data and meta data

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1. Comparison - Wet Lab vs Mutexa on Methyltransferase

Status Quo	With Mutexa
It takes **3 months** to explore one enzyme sequence	It only takes **7 days**
It costs **$200,000** to explore one enzyme sequence	It only costs **$60**
The success rate with random selection is **< 5%**	The success rate is **20% to 30%**
Collecting data from diverse sources is a **major pain point**	Our proprietary database has all the data you need, **ready for use**

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Timeline

- **EnzyHTP-1 Published [October 2022]**
- **First Enzyme HMT Optimized**[January 2023]
- **Mutexa Founded**[February 2023]
- **First AI-Based Chemical Function Released**[February 2023]
- **MVP of Mutexa Released**[June 2023]
- **GPT-Powered Mutexa Released**[December 2023]
- **Academia User License [January 2024]**
- **Commercial User License Released [May 2024]**

Yang, Z. J., Shao, Q., Jiang, Y., Jurich, C., Ran, X., Juarez, R. J., ... & Ding, N. (2023). Mutexa: a computational ecosystem for intelligent protein engineering.

https://github.com/ChemBioHTP

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1. **Why we need your help**
2.

Xinchun Ran
External CEO
Ph.D. Candidate in Chemistry
at Vanderbilt University

Dr. Zhongyue (John) Yang
CSO
Assistant Professor of Chemistry at
Vanderbilt University

We are seeking funding for:

- More customer discovery (traveling to conferences, working with technical consultants to expand our network and reach, etc.) [**50%**]
- Developing our prototype further (website, software UI/UX development, etc.) [**50%**]

Let's catalyze the future of enzymology. Together.

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Mateka

Thank you

January 2024

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